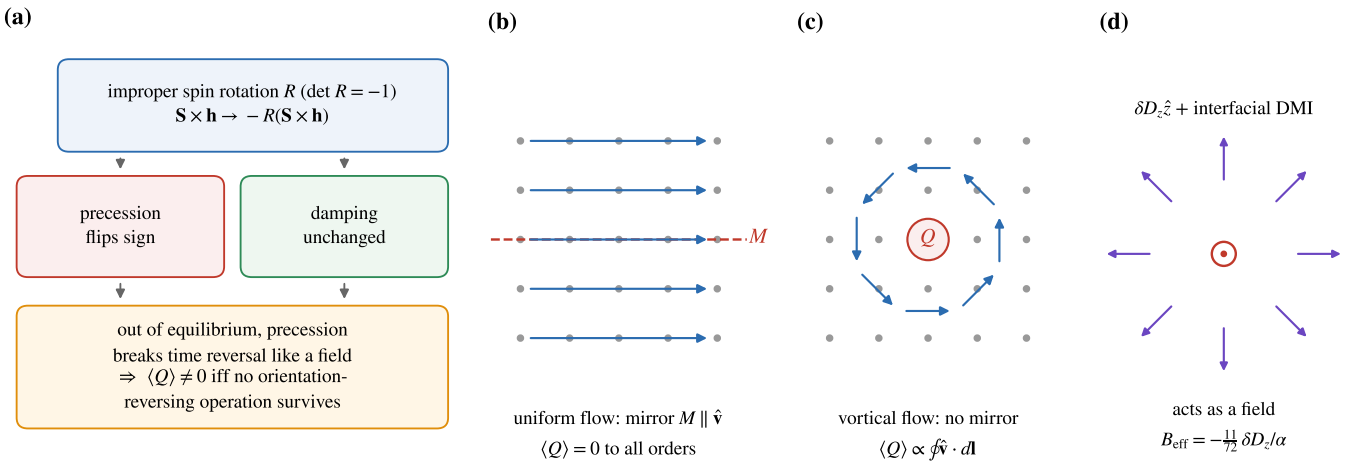
(a)
improper spin rotation R ($\det R = -1$)
$\mathbf{S} \times \mathbf{h} \to -R(\mathbf{S} \times \mathbf{h})$
precession
flips sign
damping
unchanged
out of equilibrium, precession
breaks time reversal like a field
$\Rightarrow \langle Q \rangle \neq 0$ iff no orientation-
reversing operation survives
(b)
M
uniform flow: mirror $M \parallel \hat{\mathbf{v}}$
$\langle Q \rangle = 0$ to all orders
(c)
Q
vortical flow: no mirror
$\langle Q \rangle \propto \oint \hat{\mathbf{v}} \cdot d\mathbf{l}$
(d)
$\delta D_z \hat{z}$ + interfacial DMI
acts as a field
$B_{\text{eff}} = -\frac{11}{72}\, \delta D_z / \alpha$